



MSCADMpox: A novel multi-stage classification model for effective monkeypox classification

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Abstract

Monkeypox (Mpox) is a skin disease that appeared after the COVID-19 pandemic and spread to 127 countries. Early diagnosis of Mpox according to symptoms is difficult because it is closer to measles and chickenpox symptoms, while both lesions are transmitted through respiratory droplets. However, according to skin imaging, computer-aided diagnosis systems (CADs) help diagnose the disease in the early stages. In this study, a novel model called multi-stage CAD for Monkeypox (MSCADMpox) is proposed. The proposed MSCADMpox comprises multiple stages. Firstly, a new proposed technique called RSWGAN-GP has been introduced to balance the dataset and address the overfitting issues. Secondly, features are extracted by using different handcrafted feature extraction techniques and four pre-trained models, such as Vision Transformer, VGG16, VGG19, and ResNet50. At the third stage, a new feature selection technique, MBGWO, was applied to select the most suitable features. MBGWO enhances optimal feature selection by identifying high-value frequent features and repeatedly refining the selection based on the best solution, mean, and standard deviation among all wolves. Finally, two datasets have been used, the Monkeypox Skin Lesion Dataset (MSLD) and the Monkeypox Skin Image Dataset (MSID), to ensure the effectiveness of the proposed model. To automate the diagnosis process, the proposed model uses a support vector machine (SVM) in the binary classification of the MSLD dataset and a random forest for the multiclassification in the MSID dataset that contains four skin lesions. To assess the effectiveness of the proposed model, comprehensive performance metrics such as accuracy, precision, recall, F1-score, and specificity are employed. The proposed model using the MSLD has achieved 90.54% accuracy, 91.43% precision, 88.49% recall, 90.14% F1-score, and 90.5% specificity. Moreover, the MSCADMpox has achieved 91.88% accuracy, 92.21% precision, 91.88% recall, 91.62% F1-score, and 96.93% specificity in the case of the MSID dataset.

Keywords Monkeypox (Mpox) · Generative adversarial learning gradient penalty (WGAN-GP) · Feature extraction · Grey Wolf Optimizer (GWO) · Explainable AI

Abbreviations

AI	Artificial intelligence
BN	Batch normalization
CDC	Centers for disease control and prevention
CNN	Convolutional neural network
DL	Deep learning
FC	Fully connected layers

GAN	Generative adversarial network
GLCM	Gray-level co-occurrence matrix
GWO	Grey Wolf Optimizer
HOG	Histogram of oriented gradients
LBP	Local binary patterns
LIME	Local interpretable model explanation
ML	Machine learning
Mpox	Monkeypox
MSE	Mean square error
MSID	Monkeypox skin image dataset
MSLD	Monkeypox skin lesion dataset
PSNR	Peak signal to noise ratio
RS	Residual block
SE	Squeeze and excitation
SSIM	Structural similarity index
SVM	Support vector machine

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ViT	Vision transformer
WGAN-GP	Generative adversarial learning Gradient penalty
WHO	World Health Organization
XAI	Explainable artificial intelligence

Introduction

Monkeypox (Mpox) is a rare viral disease caused by the monkeypox virus, which belongs to the same family of viruses as the variola virus, the causative agent of smallpox. Mpox is a zoonotic disease, meaning it can be transmitted from animals to humans, but it can also spread between humans. The main way that the virus infects humans is by the bite of an infected bat or primate, like a monkey [1]. Animals can transmit the virus to humans through contact with their bodily fluids, sores, or polluted materials. Transmission can occur among individuals through respiratory droplets, cutaneous lesions, or shared contaminated objects [2]. Mpox has been present since ancient times and was previously eradicated, but it has re-emerged, as announced by the World Health Organization (WHO) and the Centers for Disease Control and Prevention (CDC) in May 2022 [3]. As of November 2024, there were 117,663 confirmed Mpox cases and 263 recorded deaths, with the virus spreading over 127 countries [4].

One of the most prevalent types of illness is skin infection. Identifying illness symptoms is crucial to mitigate the spread of lesions. In most cases, symptoms of smallpox manifest between 1 and 21 days following exposure and continue for 2 to 4 weeks; however, those with compromised immune systems may experience longer symptoms. Typical symptoms as shown in Fig. 1 encompass rash, fever, headache, myalgia, dorsalgia, fatigue, and lymphadenopathy [5]. The rash typically initiates on the face and spreads to various body regions, evolving from flat lesions into fluid-filled vesicles that crust before recovery [2, 5].

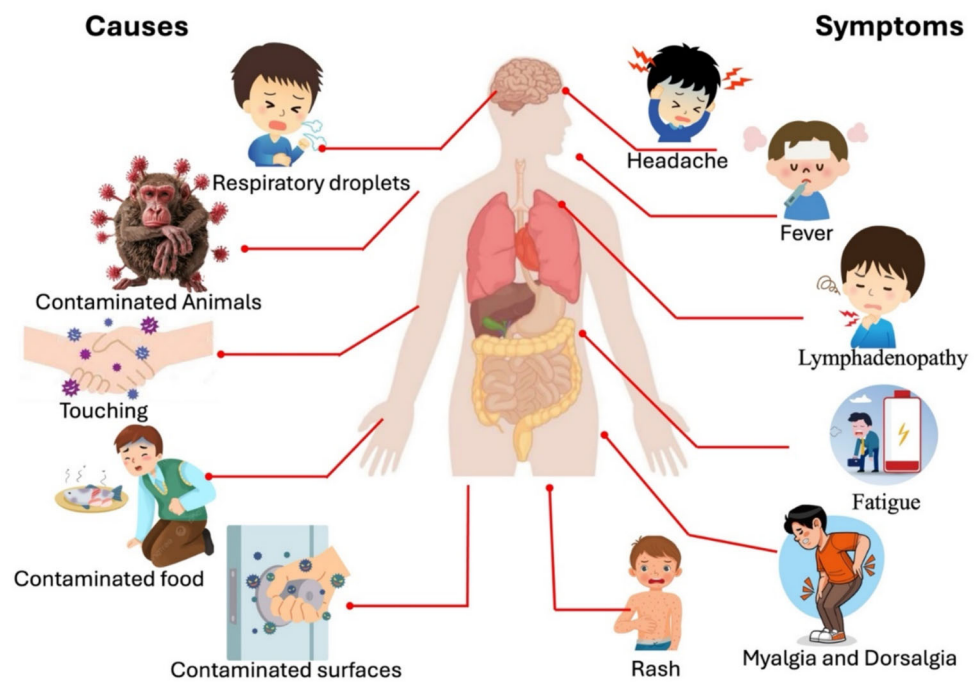
Mpox advances through multiple phases throughout roughly 2 to 4 weeks. After an incubation period of approximately 1 to 2 weeks, early symptoms manifest, including fever, headache, myalgia, lumbar pain, lymphadenopathy, chills, and fatigue. A rash often emerges 1 to 5 days following the onset of fever, frequently commencing on the face and gradually spreading to other body regions. The rash evolves through several stages, beginning as macules (flat, discolored regions), advancing to papules (elevated spots), then to vesicles (little fluid-filled blisters), followed by pustules (pus-filled lesions), and ultimately creating scabs that detach as the lesions resolve. The person is contagious from the

commencement of symptoms until all scabs have detached, and a new layer of skin has developed [5].

Over the past few years, automated computer systems that have been enhanced by artificial intelligence (AI) have shown promising results in a variety of medical applications. These systems function as knowledge-based systems that can assist in the detection of diseases and the generation of precise medical diagnoses, potentially reducing their global diffusion [6]. Researchers have increasingly concentrated on machine learning (ML) and deep learning (DL), especially convolutional neural networks (CNN), for the analysis of medical imaging and disease diagnosis [7]. DL possesses enhanced capabilities in evaluating images, including the ability to extract features, train a set of data to identify patterns, and assist with classification problems. Extracting features can be done with multiple techniques, like using handcrafted features, i.e., shape, texture, and color histogram, or using pretrained models, i.e., VGG16, ResNet, and VGG19. To reduce these features and get the most relevant features, multiple techniques are used in reduction, like filters, wrappers, embedded methods, or using metaheuristic techniques.

Although ML and DL have promising results in medical applications, their performance depends on the balance of the dataset. The generative adversarial network (GAN) was created by Goodfellow et al. [8] to produce a more balanced dataset and to make it easier to generate data without changing the information in images or applying geometric transformations. The main goal is to create synthetic data, including balancing class distributions in the dataset, reducing overfitting, and offering an alternative to traditional data augmentation methods like rotation and scaling. GAN is a type of deep learning model that can generate new data similar to realistic data to improve the performance of the classification model. Unlike basic augmentation procedures, which may harm the local position information of medical images. GAN consists of two main models: the generator, which creates new data, and the discriminator, which evaluates whether the generated data closely resembles real data [9]. Throughout the training process, the generator acquires the capability to generate progressively actual outputs in an attempt to deceive the discriminator, while the discriminator enhances its efficiency in distinguishing real data from fake data. This adversarial process produces a generator proficient at synthesizing high-quality, realistic images, sounds, or other data forms.

Recently, monkeypox and skin lesions have received a lot of attention, owing to the increasing global outbreaks and the necessity for precise diagnosis. Numerous studies seek to employ computer vision for the early detection of illnesses, representing a relatively recent development. For example, Ali et al. [10] used the MSLD dataset, which had multiple classes, and classified the dataset by using a pre-trained model, i.e., VGG16, ResNet50, DenseNet121,

Fig. 1 Causes and Symptoms of Mpox disease

MobileNetV2, EfficientNetB3, InceptionV3, and Xception. The data was augmented by using standard techniques and techniques for color space augmentation. DenseNet121 achieved the highest accuracy, with a score of 83.59%. Uysal et al. [11] acquired data from MSID for the multi-classification process, while to balance the dataset, traditional augmentation techniques were used. Their proposed model was a hybrid CNN with an LSTM model, which achieved an accuracy of 87%. For feature extraction, Maqsood et al. [6] used six modified DL models, i.e., ViT, Swin Transformer, ResNet-50, ResNet-101, EfficientNetV2, and ConvNeXt-V2. The best feature selection process used a firefly optimizer, which achieved 98% accuracy, although the data was augmented by using traditional augmentation techniques, which increased the dataset by 7919 while the original dataset was 770 images, which caused the overfitting problem and lacked diversity.

Irmak et al. [12] proposed three pre-trained models: MobileNetV2, VGG16, and VGG19 with the MSID dataset. Transfer learning was employed to enhance the performance of the models. MobileNetV2 achieved the highest accuracy at 91.38%, outperforming both VGG16 and VGG19. Sitaula et al. [13] proposed an ensemble model that combines both Xception and DenseNet-169, while they compared this proposed model with 13 different pre-trained models. The ensemble model achieved the highest accuracy compared to the other pre-trained models, with an accuracy of 87.13%. Ahsan et al. [14] utilized three modified pre-trained models: M-VGG16, M-ResNet50, M-ResNet101, and ViT. The M-ResNet50 achieved the best accuracy with 89%. LIME

is used for interpreting the model. Sahin et al. [15] developed an Android mobile application using pretrained models: ResNet18, GoogleNet, EfficientNetB0, NasnetMobile, ShuffleNet, and MobileNetV2. MobileNetV2 achieved the highest performance at 60 epochs with 91.11%. Despite getting high accuracy, it has limitations in the lower precision, specificity, and F1-score, which achieve 90% equally.

Oztel et al. [16] combined the PAD-UFES-20 and MSLD datasets, and the number of classes became 7 classes. They created an ensemble model using pre-trained DenseNet 201 and Vision Transformer. The model achieved an accuracy of 81.91% and 87.16% precision. Ahsan et al. [1] collected the dataset from Kaggle and websites; the first study has 76 images, while the second has 818 images. They used six pretrained models: VGG16, InceptionResNetV2, ResNet50, ResNet101, MobileNetV2, VGG19, and Vision Transformer. The first study achieved the best accuracy at 93% with VGG19, while the second study achieved the best accuracy at 99% with MobileNetV2. Kundu et al. [17] proposed federated learning (FL) with a deep learning model. The dataset is a combination of MSLD and MSID. MSLD consists of original images and augmented images, which combine all folders. The combined data was augmented using CycleGAN techniques to become 11,023 images; the proposed model achieved 97.9% accuracy (Table 1).

The rest of this paper is organized as follows. Sect. “Motivations and contributions” presents the motivations and contributions. Sect. “Materials” shows the material used in the proposed methodology. Sect. “The Proposed MSCADM-pox Model” shows the proposed model, MSCADMpox.

Table 1 A brief comparison between monkeypox diagnosis disease

Reference	Dataset	Models	Contribution	Cons	Accuracy
Ahsan et al. [1]	Two studies collected from Kaggle (binary)	VGG19, MobileNetV2	Trained the two studies on the six pretrained models	Limited dataset size, data quality and availability, and limited generalizability	1st study: 93%, 2nd study: 99%
Maqsood et al. [6]	MSLD	ViT + Swin transformer + ResNet 50 + ResNet 101 + EfficientNetV2 + ConvNetXt-V2	Used seven pre-trained models for extracting features and used entropy-controlled firefly for optimization selection features	The traditional augmentation techniques which lead to the augmented images are too large compared to the source data thus may cause overfitting. Where the source image 770 and the augmented data 8689 images	98%
Ali et al. [10]	MSLD	DenseNet121	Benchmark several pre-trained models: VGG16, ResNet50, DenseNet121, MobileNetV2, EfficientNetB3, InceptionV3, and Xception. The proposed model incorporates within a web application	Color space data Augmentation, transfer learning constraints	83.59%
Uysal et al. [11]	MSID	CNN + LSTM	Perform traditional augmentation techniques and combine the proposed CNN with the LSTM model	Dataset imbalance, and hybrid model complexity	87%
Irmak et al. [12]	MSID	MobileNetV2	Three pre-trained models were evaluated: MobileNetV2, VGG16, and VGG19 on the proposed dataset	Class imbalance, and potential overfitting	91.38%
Sitaula et al. [13]	Collected dataset from various sources	Xception + DenseNet-169	Perform classification of skin lesions and compare the proposed ensemble model with 13 pretrained models	Collected dataset with augmented images, The proposed model has limited accuracy	87.13%
Ahsan et al. [14]	Three types of datasets	M-ResNet50 + FL	Proposed three modified pre-trained models: VGG16, ResNet50, and ResNet10	Federated learning implementation details, and computational complexity	89%

Table 1 (continued)

Reference	Dataset	Models	Contribution	Cons	Accuracy
Sahin et al.[15]	Images gathered through the mobile device's camera	deep learning, TensorFlow	Developed an Android mobile application using pretrained models: ResNet18, GoogleNet, EfficientNetB0, NasnetMobile, ShuffleNet, and MobileNetV2	Mobile device constraints, and dataset limitations	91.11%
Oztel et al. [16]	PAD-UFES-20 + Monkeypox	Vision Transformer + Densenet 201	Combined two datasets to become 7 classes and used the ensemble model vision transformer and Densenet 201	The proposed model has a limited accuracy	81.91%
Kundu et al. [17]	Combined dataset	FL + DL + CycleGAN	Used CycleGAN for augmenting images and combined this model with FL and DL models	Computational and communication overheads, and security and privacy considerations	97.9%

Experiments and results are shown in Sect. “[Simulation Results](#)”. Finally, this paper is concluded in Sect. “[Conclusion](#)”.

Motivations and contributions

This section outlines the motivation behind implementing the proposed MSCADMpox model, followed by a summary of the key contributions made in this study.

Motivations

The study motivations can be summarized as:

- The rapid spread of monkeypox and its apparent similarity to other dermatological disorders, including measles, chickenpox, and prevalent skin lesions, pose a significant public health issue.
- Monkeypox is a communicable disease that can be spread by direct skin contact, rendering early and precise diagnostics crucial to prevent increased transmission. In many regions, especially underserved or remote areas, access to trained dermatologists or healthcare professionals is limited.
- This highlights the need for an intelligent, automated diagnostic tool that can assist in identifying monkeypox and related skin diseases from images, even in the absence of a medical expert.

- In a sufficient number of studies that integrate handcrafted features such as GLCM and LBP with deep learning-based features, therefore MSCADMpox is proposed.
- Design a hybrid model to combine and leverage both feature types for more accurate skin lesion classification. Each of these feature extraction methods generates high-dimensional features, which may contain redundant or irrelevant information that can negatively affect model performance. Therefore, to select the most informative and optimal features from this combined feature, MBGWO is introduced for optimal feature selection.

Study contributions

This paper presents a detailed model designed to assist in the detection of diseases and the identification of early-stage skin lesions. Various techniques are employed to improve the model's effectiveness. In light of the previously addressed issues, MSCADMpox is introduced as a model for diagnosing Mpox or skin lesions. Several significant contributions have been proposed in this study, which can be outlined as follows:

- 1- Employing different data augmentation techniques to balance the dataset while minimizing the risk of overfitting. Moreover, a new technique, Residual and Squeeze WGAN-GP (RS-WGAN-GP), is proposed. It has a squeeze and excitation block that improves feature separation and focuses on the most important data, as well as



Fig. 2 MSLD dataset samples

- a residual block that lets the network learn residual mappings instead of trying to learn the desired output right away. To further enhance the resolution and reduce the noise of the generated image, a bilateral filter is applied.
- 2- Utilization of a combination of deep learning features and handcrafted features. For the handcrafted features, the Gray-Level Co-occurrence Matrix (GLCM) and Local Binary Pattern (LBP) are used to identify discriminative features in an image. A Histogram of Oriented Gradients (HOG) is a statistical tool that can be used to describe the overall structure. DL features include the utilization of VGG16, VGG19, ResNet, and Vision Transformer.
 - 3- A new feature selection technique is proposed named Mean Binary Grey Wolf Optimize (MBGWO) that identifies the optimal feature set from all combined features. MBGWO is a modified version of GWO that uses a new fitness function to move the positions of alpha, beta, and delta wolves from continuous space to discrete space. Improving optimal feature selection involves identifying features that frequently appear with an average value greater than 0.8 and comparing them to the initial optimal features to find common features. This process is implemented by calculating the mean and standard deviation for each feature after acquiring the best feature matrix for whole wolves. Every iteration of calculating optimal feature selection is done by using two methods: one is based on the best solution, and the other is by applying mean and standard deviation. The fitness value is then computed for the optimal solution to determine whether it exceeds the fitness value of the current optimal solution. Finally, the fitness value is calculated to see if it surpasses the fitness value of the optimal solution. Then, select the feature that achieves the best fitness value.

- 4- To understand the behavior of the proposed model's output, explainable artificial intelligence (XAI) is used. LIME is used to provide the most important features that are more influential in the classification process.

Materials

The classification experiment for Mpox utilized two datasets, which are the Monkeypox Skin Lesion Dataset (MSLD) [18] and the Monkeypox Skin Image Dataset (MSID) [19]. The MSLD dataset comprises 228 images in total, with 102 categorized under the 'Monkeypox' class and the remaining 126 classified as 'Others,' which includes non-monkeypox cases such as chickenpox, measles, and uninfected skin, as shown in Fig. 2. The image resolution is 224×224 pixels in JPG format. The MSID dataset consists of four classes illustrating four different skin conditions: monkeypox, chickenpox, measles, and uninfected skin, as shown in Fig. 3. The dataset includes a total of 279 instances of monkeypox, 107 instances of chickenpox, 91 instances of measles, and 293 instances of uninfected skin, respectively. The dataset consists of a total of 770 images. The images within the dataset were formatted as PNG. Table 2 provides the distribution of images across different classes for both datasets, MLD and MSID.

The proposed MSCADMpox model

The proposed model for classification and identification of monkeypox, namely MSCADMpox, consists of four main stages, as shown in Fig. 4. Where the first stage has applied the proposed RSWGAN, which is responsible for generating



Fig. 3 MSID dataset samples

Table 2 Dataset Description

Dataset	Classes		Total
MSLD	Others	126	228
	Monkey Pox	102	
MSID	Chickenpox	107	770
	Measles	91	
	Monkeypox	279	
	Normal	293	

synthetic data. Then the original data has been combined with augmented data for the feature extraction process, which uses four pre-trained models and traditional feature extraction. The third stage involves applying the proposed MB-GWO models to select the optimal features. The final stage is responsible for classification and measuring the performance by using performance metrics.

The proposed RS-WGAN-GP model

GANs have gained significant attention as a powerful generative model, allowing researchers to produce high-quality synthetic data as an alternative to traditional augmentation techniques such as rotation and scaling. The loss function in GAN uses Jensen-Shannon Divergence (JS-Divergence) [20] to measure the disparity between the distribution of generated data and the actual data distribution. The primary goal of the loss function in GANs is to approximate the probability distribution of both the generator and discriminator. This creates a competitive dynamic between the generator and discriminator, driving the generator to produce synthetic data that closely resembles real data. However, despite the effectiveness of GANs, they suffer from challenges such as mode collapse and training instability due to vanishing or exploding gradients [21]. The loss function L_D that defines the objective of a GAN can be formulated as follows:

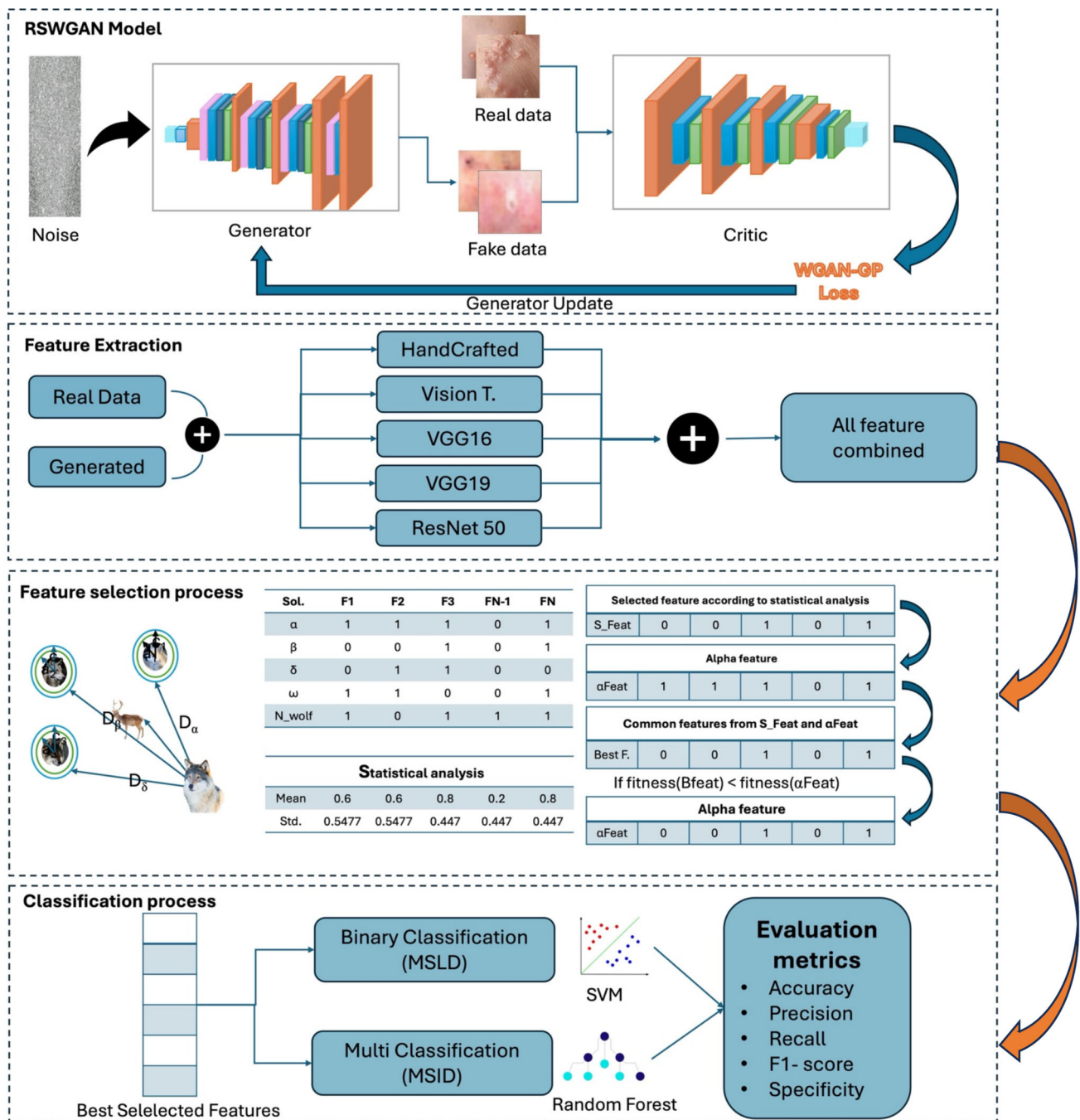


Fig. 4 The Proposed model MSCADMpox

$$L_D = E_{x \sim P_r} [\log D(x)] + E_{\hat{x} \sim P_g} [\log(1 - D(\hat{x}))] \quad (1)$$

where P_r represents the real data distribution and initial random noise for the discriminator, which P_g is the generator's distribution. x represents for real data samples and $\hat{x}$ is for generated data samples. The first term represents the probability of real samples being classified as real, and the second

term represents the expected probability of generated samples being classified as fake. $D(x)$ Present the output for the discriminator on the real data, where $D(\hat{x})$ is the output for the discriminator for generated data. The generator aims to maximize $D(\hat{x})$, while the discriminator aims to minimize it, thus enhancing its capacity to differentiate between real and generated data.

To mitigate the vanishing gradient issue, Arjovsky et al. [22] proposed WGAN, which used the Wasserstein distance

to measure the difference between the probability distributions of real and generated samples, improving the loss function and enhancing weight updates [23]. In WGAN, the discriminator is substituted with a critic that assesses the similarity between generated data and real data. To implement the Lipschitz restriction necessary for stable training, WGAN initially utilized weight clipping. This strategy frequently resulted in unfavorable outcomes, including inadequate convergence and limited model capacity. To address these constraints, Gulrajani et al. proposed WGAN-GP [24], which substitutes weight clipping with a gradient penalty, resulting in more stable and efficient training. The objective loss function WGAN-GP is formulated as follows:

$$L = E_{\hat{X} \sim P_g} [D(\hat{X})] - E_{x \sim P_r} [D(x)] + \lambda E_{\hat{x} \sim P_{\hat{x}}} [(\|\nabla_{\hat{x}} D(\hat{x})\|_2 - 1)^2] \quad (2)$$

$E_{\hat{X} \sim P_g} [D(\hat{X})] - E_{x \sim P_r} [D(x)]$ is the critic loss, where x represents for real data and P_r for real data distribution. $\hat{X}$ represents the generated data and P_g is for the distribution of generated data. Where the second is for gradient penalty loss. $\hat{x}$ denotes an intermediate sample between a real and a generated sample. $P_{\hat{x}}$ represents the random sample between pairs of points from the generator and discriminator distributions. λ is a hyperparameter to control the gradient penalty.

For the above models, WGAN-GP is used to enhance the generated data by using the proposed RS-WGAN-GP. The structure consists of two main models: a generator and a discriminator. The generator consists of five deconvolution layers with decreasing filter values (512, 256, 128, 64, 3) with kernel size (4, 4), which is the main purpose of sampling and synthesizing images, gradually constructing the image from a low-dimensional latent space as shown in Fig. 5. After the first four deconvolution batch normalization (BN) is applied to normalize activations. This ensures that the output of each layer maintains a mean close to zero and a standard deviation around one to hasten training and enhance the model stability. A LeakyReLU activation function is then applied to improve gradient flow and enhance performance in the learning process. Following this, a modified squeeze and excitation block is added to reduce the dimensionality of the input and to focus on the most important dependence and relevant information. Finally, a residual block is added for the next section.

For the discriminator model, as shown in Fig. 6, the discriminator model comprises five convolution layers that extract features and downsample data to distinguish between real and fake images. The sequence of convolution layers with increasing filters is 64, 128, 256, and 512, and the kernel size is 4; each of them is followed by the LeakyReLU activation function, and the residual block is used to preserve the information and enable deeper extracted features. The last

convolution layer uses single filters followed by a flattened layer to convert the output to a 1D vector. Next, a dense layer with a single neuron output generates the final prediction, enabling the distinction between real and synthetic data.

RS-WGAN-GP is designed to generate images with a size of (64, 64). The number of epochs is 1000, where the discriminator is trained 3 times for each training step of the generator. During training, the Adam optimizer is used with a dynamic learning rate that starts at $1e-4$ and a decay factor of 1.00004 to make sure that the rate slowly goes down for stable training. To create realistic visuals, the generator uses random noise vectors as input that have been sampled from a normal distribution with a dimension of (64, 100). These vectors are then transformed through a number of transposed convolutional layers. The discriminator evaluates the generated image by using some of the convolution and residual layers to enhance the generator's efficiency. The lambda factor, with a value of 10, controls the gradient penalty factor to stabilize the training and prevent the model collapse problem.

Some limitations appear in the used dataset, as it doesn't have the same object or the same structure, but it consists of different images from different parts of the body, which means the dataset consists of legs, arms, bodies, faces, hands, full bodies, and different nationalities of humans, which makes the generator take more time to train the structure and makes the general process difficult. Therefore, to solve this problem and achieve sufficient processing, the dataset should be split into three different categories: legs and hands, faces, and body parts. In addition, the generator has been trained with three sets separately. Finally, after the image generation process has been finished, the resulting images are compared to the original dataset. The resulting images have some blurring and are noisy, so a bilateral filter is applied to enhance the quality of the newly generated images. Seventeen images are selected from the newly generated set and added into the Mpox class to achieve balance within the MLSD dataset. As shown in Fig. 7, it presents samples of generated data measured by Peak Signal to Noise Ratio (PSNR) and Structural Similarity Index Measure (SSIM).

Squeeze and excitation

Figure 8 illustrates the Squeeze and Excitation (SE) blocks. It is used to facilitate feature recalibration within the network. This method allows the network to utilize global information to selectively highlight informative features and diminish less advantageous functions [25]. The initial procedure involves a squeeze that reduces the spatial dimensions of the input tensor of size (H, W, C) through the use of a global average pooling layer. As shown in Fig. 8, the input for the SE block is a feature map with dimensions (H, W, C), which is processed through a global average pooling layer to reduce

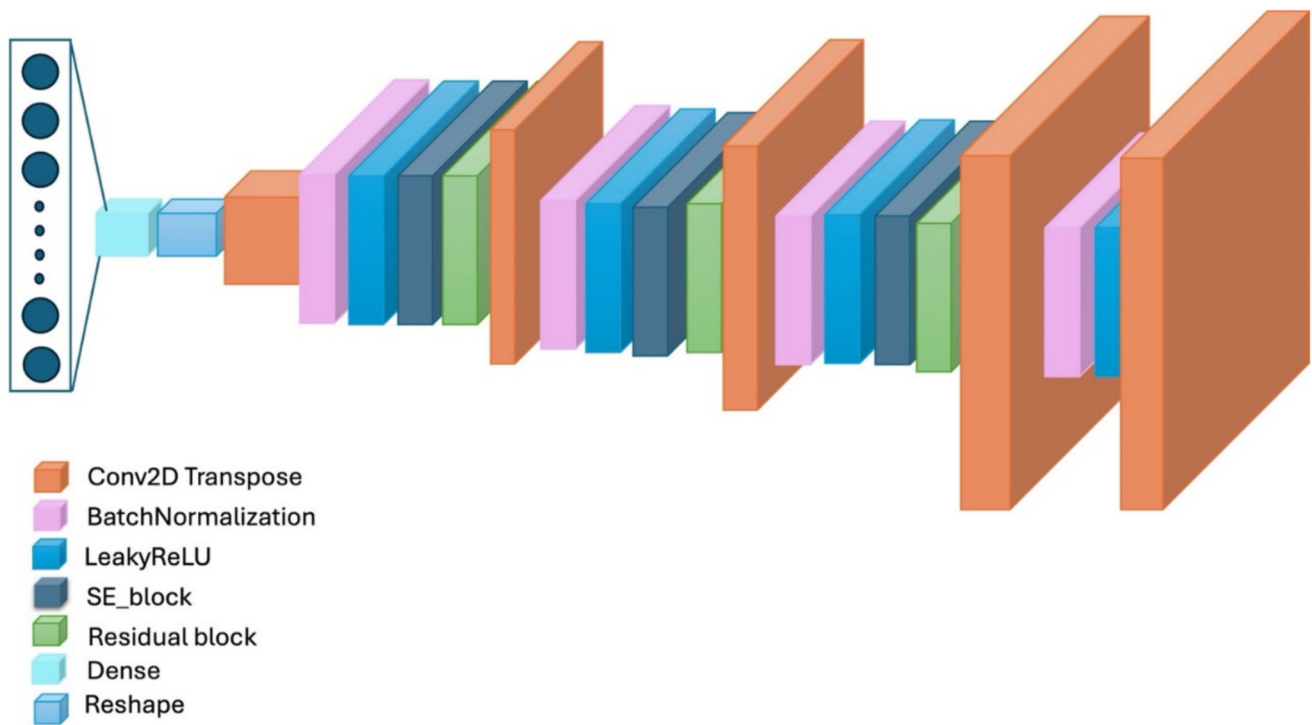


Fig. 5 Architecture of the proposed generator model

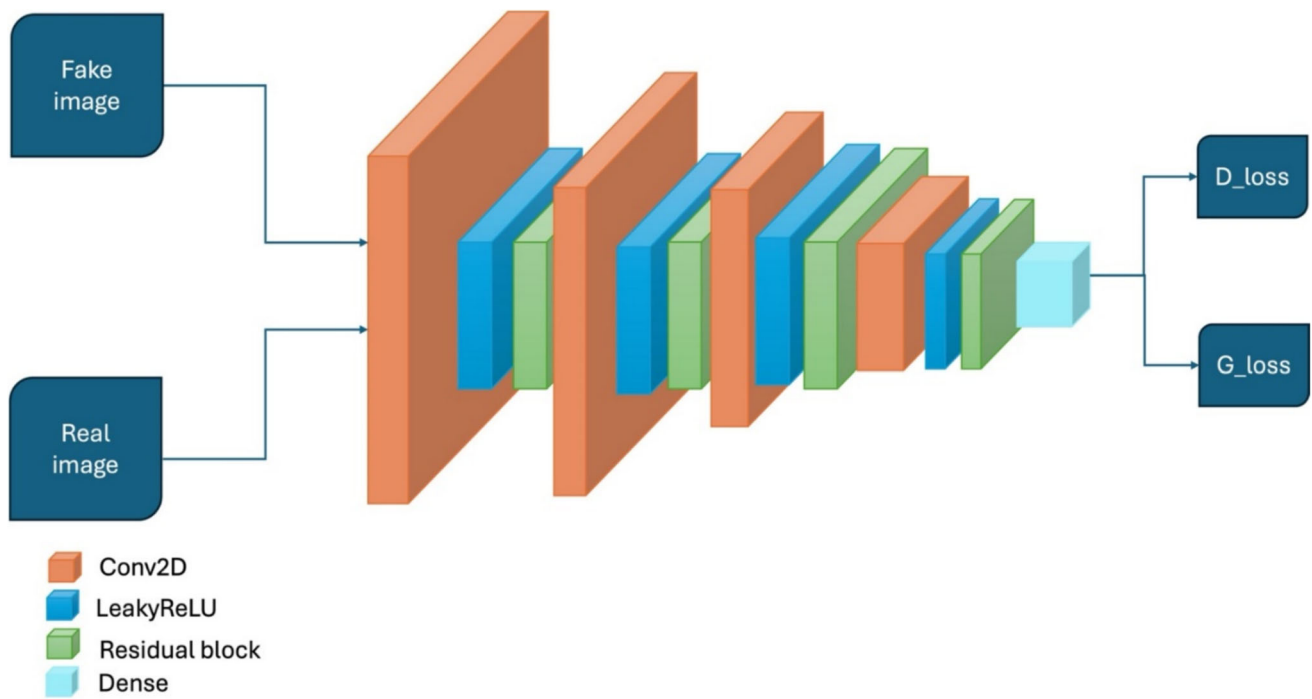


Fig. 6 Architecture of the proposed discriminator model



Fig. 7 Output samples from RS-WGAN-GP model

the spatial dimensions. The next phase is the excitation process, which consists of two fully connected layers (FC). The first FC is used to reduce the dimensionality of the input, which is the input size from the squeeze process (batch_size, C). This reduction is achieved by a reduction ratio of 16, employing the ReLU activation function to produce a feature map of dimensions (batch_size, C/16). The second fully connected layer is employed to revert the input channel's dimensions to (batch_size, C) utilizing the sigmoid function. During the recalibration phase, element-wise multiplication is performed between the input tensor of dimensions (H, W, C) and the output of the second fully connected layer, which has dimensions (1, 1, C). This multiplication allows the model to enhance the feature selection process by prioritizing the most essential features and diminishing the less informative ones.

Residual block

A residual block (RS) is used after the SE block, which emphasizes the important feature. The RS block allows the model to learn additional spatial patterns and hierarchical features while preserving enhanced channel-wise importance. Additionally, RS helps the model to reduce computation complexity, and the skip connection in the RS block stabilizes the training process by maintaining identity mappings, improving gradient flow, and facilitating deeper feature extraction without losing the recalibration benefits introduced by the SE block. The combination of these

two methods improves both channel-wise and spatial feature representations, making feature maps more stable and expressive. As shown in Fig. 9, residual blocks consist of four layers. The first layer is a convolution layer, which allows for feature extraction from the input with a reduction in the dimensionality. It is followed by the BN and LeakyReLU objective function. The last layer is the convolution layer, which is combined with a skip connection.

The feature extraction stage

Feature extraction is the process of transforming the image information into numerical values that can be used in the classification process. These values are representative of the texture, color, and shape of images. The feature extraction process can be handled by using two methods: handcrafted or deep learning techniques. In the proposed model, both methods are employed: handcrafted features and various pre-trained models for the extraction process.

Handcrafted feature

Handcrafted features are extracted by using the Gray-Level Co-occurrence Matrix (GLCM), Local Binary Patterns (LBP), and Histogram of Oriented Gradients (HOG) techniques. For extracting shape features, the HOG is used to extract shape features like edges or the structure of an object. Texture features are extracted by combining two techniques: LBP and GLCM. LBP texture features are derived from local

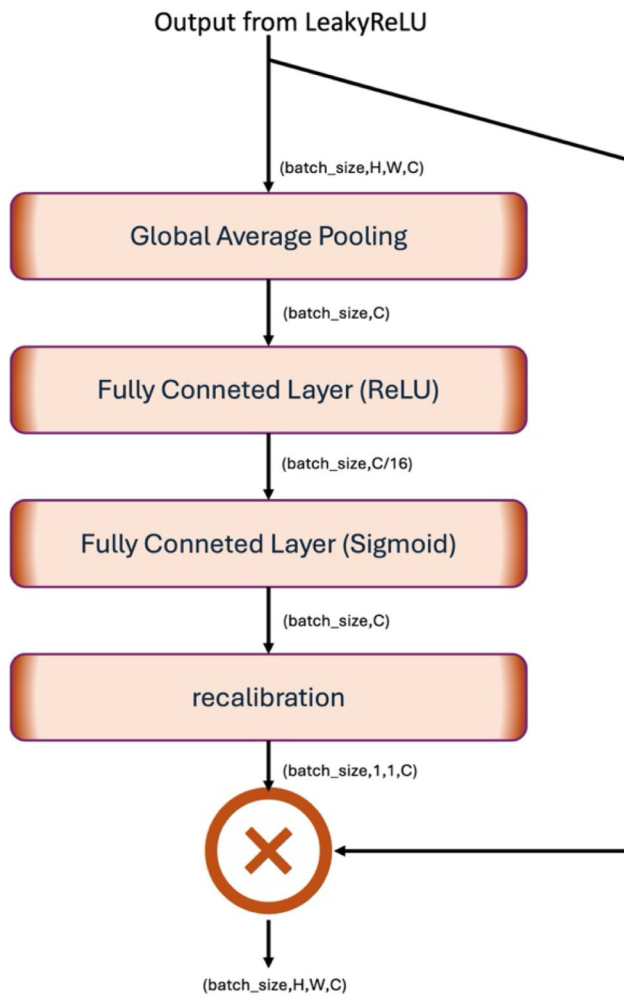


Fig. 8 Squeeze and excitation block

attributes that compare the pixel values of boundary pixels inside a window to a central reference pixel, obtained through a weighted combination across many orientations [26]. Where GLCM provides a global texture characteristic that examines the spatial correlation between every pixel. GLCM is a matrix that characterizes the joint probability of two measurements between pixel sets, based on a specified separation. Each cell denotes the frequency of co-occurrence for a specific pair of gray levels at a specific offset and direction. The co-occurrence matrix is formed in four specific orientations: 0° , 45° , 90° , and 135° [27]. GLCM has several texture features that focus on different parameters of an object. Four texture features are used: energy, correlation, contrast, and homogeneity.

In skin lesions, the energy feature can distinguish between smooth and infected areas in the skin, as calculated in (Eq. 3). A high energy value indicates a uniform texture with minimal variations in intensity levels. Conversely, the low value indicates there are skin lesions. Correlation is calculated as in

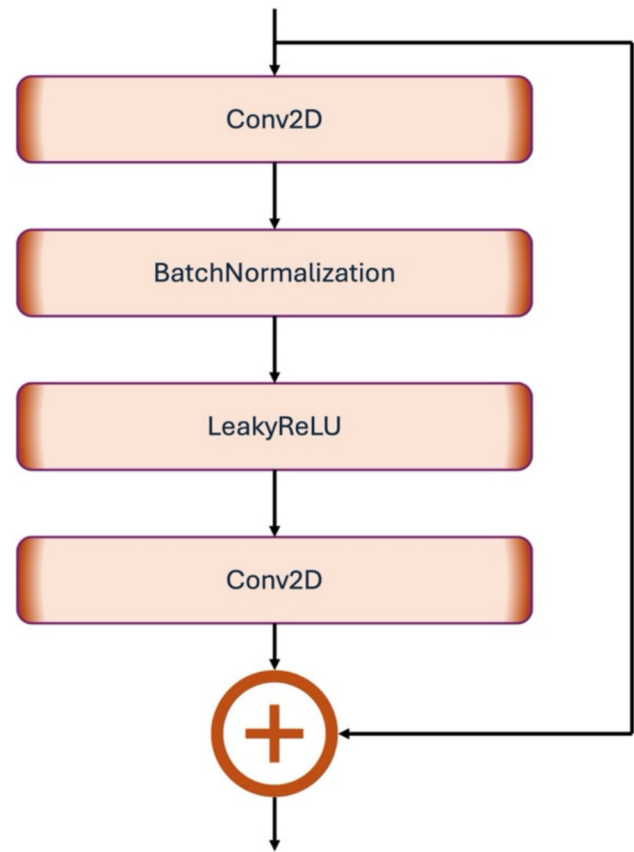


Fig. 9 Residual block

(Eq. 4), which represents the linear correlation between pairs of pixels. For contrast features, which measure the intensity relation between neighborhood pixels, it can be calculated as in (Eq. 5). The high-intensity value shows that there are lesions, which indicate there are differences between infected and non-infected regions. Homogeneity determines the closeness of the element distribution in an image to evaluate the texture of the images, as calculated in (Eq. 6). The lower value of the homogeneity indicates that the texture is more rough, which indicates the skin lesions. After extracting all of these features, they combine them together to get the final size of 210 features.

$$Energy = \sum_{x=0}^{m-1} \sum_{y=0}^{n-1} f^2(x, y) \quad (3)$$

$$Correlation = \sum_{x=0}^{m-1} \sum_{y=0}^{n-1} \frac{(x - \mu_x)(y - \mu_y)f(x, y)}{\sigma_x \sigma_y} \quad (4)$$

$$Contrast = \sum_{x=0}^{m-1} \sum_{y=0}^{n-1} (x - y)^2 f(x, y) \quad (5)$$

$$\text{Homogeneity} = \sum_{x=0}^{m-1} \sum_{y=0}^{n-1} \frac{1}{1 + (x + y)^2} f(x, y) \quad (6)$$

x and y represent for the coordinates of pixels in the image. The terms m and n are the height and width dimensions respectively of the image. μ_x is mean of pixel intensities along x -axis and μ_y mean of pixel intensity along y -axis. σ_x and σ_y represent the standard deviation of pixel intensities horizontal and vertical direction.

The deep learning features

For the importance of ML and DL in recent years. DL and ML have been used in many diseases' detection and classification problems. According to the proposed model, a traditional ML technique is used for the classification of skin lesion diseases. Given the significance of the feature extraction process for disease classification, it is aimed to enhance its robustness to achieve optimal quality; thus, DL models are employed to extract additional features. In DL, feature extraction includes deriving meaningful representations from images using pre-trained convolutional neural networks (CNNs). In this study, four popular deep learning architectures, i.e., VGG16, VGG19, Vision Transformer, and ResNet50, are used as feature extractors. DL models involve two main processes, which are classification and feature extraction, while the feature extraction process is done by using the convolution layer without applying the fully connected layer.

The VGG16, VGG19, and ResNet50 models exclude the top layer, which is a fully connected layer responsible for classification. These models are pre-trained on the ImageNet dataset, using an input shape of (224, 224, 3). The feature count retrieved from VGG16 and VGG19 is 512 for each model. ResNet50, with 48 convolutional layers, extracts 2048 features. The Vision Transformer is a DL model distinct from the CNN model, as it utilizes the transformer architecture. This model employs a vision transformer pre-trained on ImageNet, comprising 21 classes, with an input image size of (224, 224) divided into 16×16 patches. The vision transformer possesses 768 features. The pre-trained model comprises a total of 3840 features.

Feature combination

The system integrates all extracted features, including handcrafted features, VGG16, VGG19, Vision Transformer, and ResNet50, utilizing the concatenate function that processes the features as arrays. The combination sequence initiates with handcrafted features, followed by integration with vision features, then VGG16 features, subsequently VGG19 features, and finally ResNet50 features.

The proposed mean binary Grey Wolf Optimizer (MB-GWO)

According to the previous section, the extracted features have huge dimensions, which will have high computational costs in the classification process. The dataset's issue extends beyond its huge dimensionality to include unnecessary and redundant features. Moreover, the gathered datasets may contain significant noise. So, feature selection is a needed process to select the optimal features that will be used. In this study, a new optimization technique, called MB-GWO, presents a new wrapper method, which necessitates training a learning model for each subset during the feature selection process. The proposed algorithm is MB-GWO based on the Grey Wolf Optimizer (GWO). The GWO is an innovative meta-heuristic method inspired by the hunting behavior and social leadership of grey wolves, which are exceptional predators with advanced abilities in capturing their prey [28]. GWO has levels for wolves; each of them is responsible for a specific task. To search for prey, GWO has a hierarchical mechanism in which each wolf performs specific tasks to get optimal solutions [29, 30].

The hierarchy is represented in four levels: alpha (α), beta (β), delta (δ), and omega (ω). Alpha is responsible for making decisions in hunting, sleeping, and other actions, as he is the leader of all wolves, while Beta helps Alpha in their decision [31]. The first optimal solution candidates α , β , and δ , respectively. The wolves update their position to circulate the prey for hunting according to the following mathematics:

$$\vec{D} = |\vec{C} \cdot \vec{X}_p - \vec{X}(t)| \quad (7)$$

$$\vec{X}(t+1) = |\vec{X}_p - \vec{A} \cdot \vec{D}| \quad (8)$$

where $\vec{D}$ is represents the distance between wolves and the prey, where $\vec{X}_p$ and $\vec{X}(t)$ is the position of the prey vector and grey wolf respectively. $\vec{A}$ and $\vec{C}$ are coefficient vector which the value is calculated according to the following equations:

$$\vec{A} = |2 \cdot \vec{a} \cdot \vec{r}_1 - \vec{a}| \quad (9)$$

$$\vec{C} = 2 \cdot \vec{r}_2 \quad (10)$$

where $\vec{r}_1$ and $\vec{r}_2$ are random values between 0 and 1. $\vec{a}$ is a value that decrease linearly from 2 to 0 according to the following equation:

$$\vec{a} = 2 - t \cdot \frac{2}{\text{maxIter}} \quad (11)$$

The *maxIter* denotes the maximum number of iterations in the technique. The current iteration value is denoted by t , with each iteration value of $\vec{a}$ is decreasing linearly to balance exploration and exploitation. As the value of $\vec{a}$ approaches 2, the wolves focus on extensively exploring the search region; conversely, as it nears 0, the wolves concentrate on exploitation to pursue the optimal solution. Therefore, for each potential candidate in the search, the following set of equations is executed continuously to mimic the hunting mechanism of grey wolves:

$$\vec{X}(t+1) = \frac{\vec{X}_1 + \vec{X}_2 + \vec{X}_3}{3} \quad (12)$$

$$\vec{X}_1 = \left| \vec{X}_\alpha - \vec{A}_1 \cdot \vec{D}_\alpha \right|, \vec{D}_\alpha = \left| \vec{C} \cdot \vec{X}_\alpha - \vec{X} \right| \quad (13)$$

$$\vec{X}_2 = \left| \vec{X}_\beta - \vec{A}_2 \cdot \vec{D}_\beta \right|, \vec{D}_\beta = \left| \vec{C}_2 \cdot \vec{X}_\beta - \vec{X} \right| \quad (14)$$

$$\vec{X}_3 = \left| \vec{X}_\delta - \vec{A}_3 \cdot \vec{D}_\delta \right|, \vec{D}_\delta = \left| \vec{C}_3 \cdot \vec{X}_\delta - \vec{X} \right| \quad (15)$$

The proposed MB-GWO is shown in Fig. 10 for enhancing the feature selection process. Where the fitness function is used in the evaluation process, which helps in improving the learning process in reaching the optimal solution, as shown in Eq. 16. As shown in the equation of the fitness function, the classification error rate is required, which is calculated by using a support vector machine (SVM). The SVM measures the accuracy of the best solution candidate using GWO equations and then calculates the error rate.

$$fitness = \alpha * E + \beta \cdot \left(\frac{|S|}{|T|} \right) \quad (16)$$

where E is a classification error rate calculated from SVM. α and β are weight parameters, where the value of α is 0.99 and $\beta = 1 - \alpha$. $|S|$ denotes the number of selected parameters and $|T|$ is the number of total parameters.

The number of wolves in the proposed model is 10 wolves, with 30 iterations. In one iteration, the fitness function is updated according to the first best value of fitness value to update α, β , and δ wolves. For every wolf, the positions of the wolves are updated according to the Eqs. (7–12). After updating the wolves' positions, the fitness function is calculated for each wolf, and then they are sorted in an ascending order, deciding the best wolves' positions and the best fitness value. After each iteration, a new idea is suggested to improve the feature selection process. Once all wolves have been updated, the effectiveness of the features is estimated across all wolves, indicating which features are most frequently selected among the 10 wolves. The mean and standard deviation are calculated to measure the significance of each feature, as shown in Fig. 10. Then select the features that

have a mean value larger than 0.8 with sufficient variability. After performing the statistical analysis and selecting these features, the chosen significant ones are compared with those already selected by the alpha wolf, representing the optimal feature subset identified thus far. The intersection of significant features and those chosen by the alpha wolf is identified, and these overlapping features are integrated into a binary vector representation. Then the selected feature efficiency is calculated by using the objective function, which calculates the fitness function. If the fitness of the statistical analysis is lower than the fitness of the alpha fitness, it indicates it has better features; the alpha wolf feature is updated accordingly. Otherwise, the alpha wolves remain as they are. To optimize the feature subset for the given job, this iterative refinement selects the most important characteristics while minimizing redundancy.

Skin lesions classification

After extracting the features and selecting the best features based on the proposed MB-GWO, the classification process is ready to be performed with five suggestive classifiers: Random Forest (RF), Support Vector Machine (SVM), Naïve Bayes (NB), K-Nearest Neighbor (KNN), and Logistic Regression (LR). Table 3 presents the parameters of the classifiers that are used in the proposed model. In the proposed model, according to the previous stages, it is not necessary to use a complicated classifier, so in binary classification, SVM has been used. The SVM can treat high-dimensional data and find the optimal decision between different classes. The Random Forest classifier has achieved the highest accuracy in the multi-classification process.

Random forest (RF)

Random Forest is an ensemble method comprising decision trees, each trained on a bootstrap sample of the dataset and a random selection of features. Predictions are determined through majority voting among all trees. The RF classifier was trained using the Gini impurity as the splitting criterion, which measures the degree of class homogeneity within the nodes [32].

$$Gini(D) = 1 - \sum_{j=1}^k p_j^2 \quad (17)$$

Equation (17), D represents the dataset of the current node, k referring to the total number of classes in dataset, p_j is the probability of samples in the dataset.

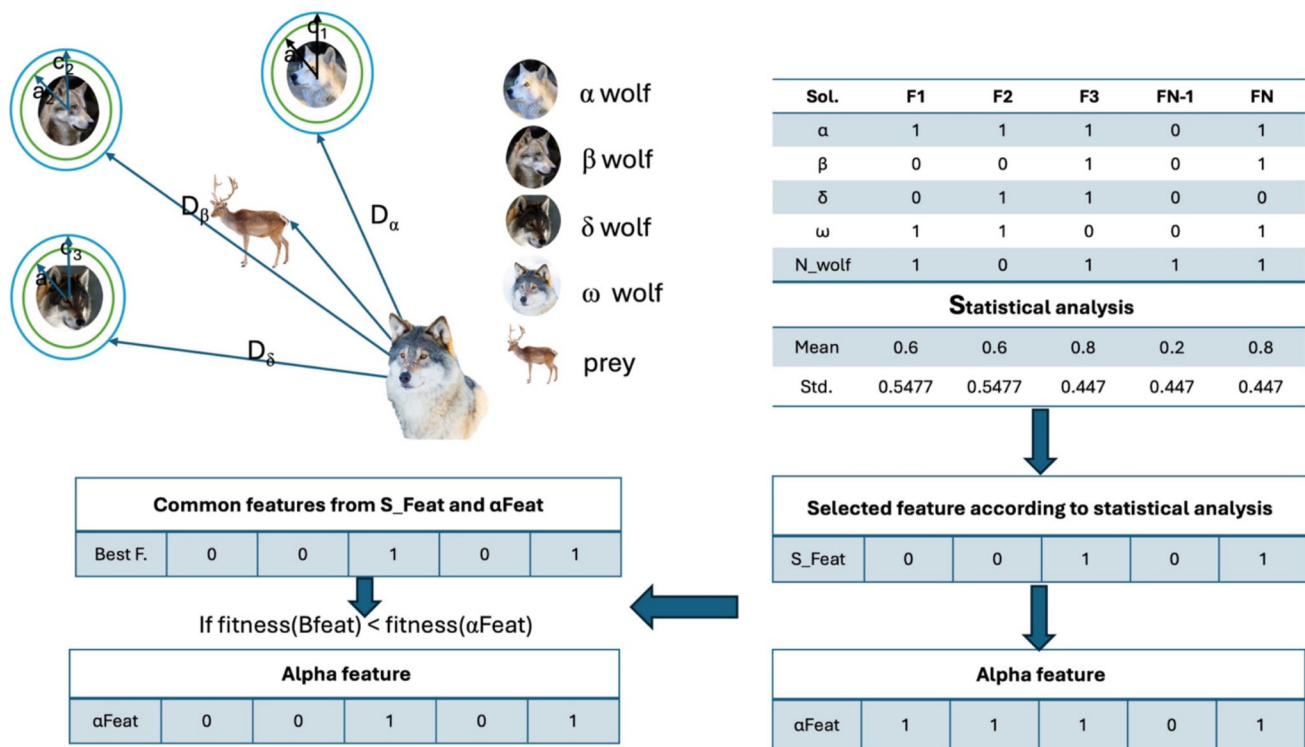


Fig. 10 The proposed MB-GWO model

Table 3 Summary of the optimizer's parameters

Classifier	Parameters
Random Forest	Random_state = 0, n_estimators = 100,
Support vector machine	Random_state = 0, probability = True
Naïve Bayes	var_smoothing = 1e-9(using default)
K-Nearest Neighbor	n_neighbors = 5, metric = "minkowski", Euclidean distance = 2
Logistic Regression	max_iter = 1000, solver = saga, C = 10

Support vector machine (SVM)

In this study, the classification model SVM approach using a Radial Basis Function (RBF), kernel is implemented via the SVC class in the scikit-learn library. The SVM is a supervised learning model that aims to identify an optimal hyperplane that maximizes the margin between data points of different classes [33].

$$\min_{f, \xi} \frac{1}{2} \|f\|^2 + C \sum_{i=1}^l \xi_i \quad (18)$$

Equation (18), C is a regularization parameter that governs the balance between margin maximization and the penalty for misclassifications, where ξ_i denotes slack variables that indicate the extent of margin violation by each data point, f refers to the weight vector.

To deal with non-linearly separable data, it is necessary to transform the data into a higher-dimensional space utilizing a kernel function K , without explicit computation. Thus, the summing terms in the dual formulation in Eq. (19) indicate the impact of each pair of training points on the decision boundary, modulated by their labels and learned coefficients (α -values). These parameters are essential for determining the appropriate separation hyperplane in the transformed space [34].

$$\min_{\alpha_i} \frac{1}{2} \sum_{i=1}^l \sum_{j=1}^l \alpha_i \alpha_j y_i y_j K(x_i, x_j) - \sum_{i=1}^l \alpha_i \quad (19)$$

Where α_i is the dual variables associated with the training sample, y_i is the class label for the training sample, $K(x_i, x_j)$ is the kernel function that computes the similarity between data points x_i and x_j .

Naïve Bayes (NB)

The Gaussian Naive Bayes classifier was utilized for its simplicity and probabilistic basis. It proposes that features are conditionally independent given the class and that each feature corresponds to a Gaussian distribution [35]. The classification prediction is derived by utilizing Bayes' theorem as in Eq. (20):

$$y = \operatorname{argmax}_y P(y) \cdot \prod_{j=1}^n P(x_j|y) \quad (20)$$

where the probability $P(x_j|y)$ is represent as a Gaussian distribution characterized by class-specific means and variances, $P(y)$ is the prior probability of class y , and n is total number of features.

K-Nearest Neighbor (KNN)

The KNN algorithm identifies optimal neighboring pixel regions around key points obtained through feature extraction and dimensionality reduction [36]. It outputs facial feature vectors along with their spatial coordinates extracted from the input face image. These descriptors are generated from the local pixel patterns surrounding each key point, where each point acts as the center of a distinct region of interest. KNN then classifies based on similarity, typically measured using the Euclidean distance, which is calculated as follows:

$$d(x, x_i) = \sqrt{\sum_{j=1}^n (x_j - x_{ij})^2} \quad (21)$$

In Eq. (21), $d(x, x_i)$ refers to the Euclidean distance between the query point x and the training point x_i ; n denotes the number of features in the feature vector, while x_j and x_{ij} represent the j -th feature of the test sample and training sample, respectively.

Logistic regression (LR)

Logistic Regression is a probabilistic linear classifier that assesses the probability of class association by utilizing a sigmoid activation function on a linear aggregation of input features [37]. It can be defined in the following equation Eq. (22):

$$p(x) = \frac{1}{1 + e^{-(w^T x + b)}} \quad (22)$$

As a training step, minimizing the binary cross-entropy loss gets the model closer to producing accurate predictions for labeled data. where $x \in \mathbb{R}^n$ is refer to the input features

and n to the number of features, w is the weight vector, b is the bias value.

Explainable artificial intelligence (XAI) algorithm

One of the major challenges in the usage of AI models is the prediction of the model output or having an explanation or a certain reason for making some decisions. Therefore, XAI was discovered to illustrate this decision. An explanation of the decision is so crucial for understanding the behavior of the model and justifying decisions. XAI is an extension of AI models that enhances them through interpretability and explainability [38]. The Local Interpretable Model-agnostic Explanation (LIME) is used to explain the predictions of ML models and make complex models easier to understand. In the proposed model, the LIME process helps to understand the effectiveness of the features selected and which are more influential in the classification process.

Simulation results

This section evaluates the performance of the proposed model, which consists of two separate procedures. The first one applies the MSCADMpox for binary classification that deals with the MSLD dataset, while the second procedure focuses on classifying the MSID dataset, which comprises four categories: chickenpox, measles, Mpox, and normal. The subsequent subsections evaluate the proposed model. In the generator process, the generated images are evaluated by using PSNR and SSIM metrics. Random Forest, SVM, Naïve Bayes, KNN, and logistic regression are used for the classification process evaluation. Performance evaluation is calculated based on five metrics: accuracy, precision, recall, F1-score, and specificity. The proposed methodology has been performed on a MacBook Air M1, using Anaconda as the package manager. The proposed model utilizes the Spyder IDE to compose and evaluate Python code. Furthermore, utilize TensorFlow and Keras for tasks involving learning and classification using Python 3.10.

Classification of MSLD dataset

The proposed model is trained on the MSLD dataset, which focuses on binary classification. To balance the dataset, RS-WGAN-GP is used. In this proposed model, the data is split into categories to make the generated data more sufficient and accurate. The dataset is divided based on different parts of the body. The MSLD dataset consists of two folders: one containing the original data, which has already been utilized, and another containing the augmented data. GAN algorithms require a large amount of training data to learn effectively, so data are collected from the original dataset along with some

Table 4 Comparison of the performance of three models of WGAN-GP

Model	PSNR	SSIM
WGAN-GP	23.12	0.56
SE-WGAN-GP	25.75	0.59
RS-WGAN-GP (the proposed)	26.33	0.61

samples from the augmented folder. The data is split into body, face, and leg categories, containing 358, 107, and 185 images, respectively. The model is trained for 1800 epochs, and after the training process, 64 images are generated. The generated images contain some noise and blurriness, so a bilateral filter is applied to enhance the synthetic data. The quality of the generated data is evaluated using PSNR and SSIM.

PSNR is a metric that defines the quality of signal reconstruction. Higher values of PSNR indicate a high quality of generated images. Where the PSNR is larger than 20 dB, that means it has good quality. SSIM is a metric for assessing the similarity between two images by comparing the structure and contrast. The SSIM value is in the range between 0 and 1, where it is close to 1, which means a high-quality image. PSNR and SSIM are calculated as shown in Eqs. (23) and (24).

$$PSNR = 10 \cdot \log_{10} \left(\frac{MAX_I^2}{MSE} \right) \quad (23)$$

$$SSIM(x, y) = \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)} \quad (24)$$

The term MAX refers to the maximum value of a pixel in an image, while MSE represents the mean square error between the original and generated images. In Eq. 14, μ_x and μ_y is the mean of images x and y respectively; σ_x , σ_y are the variance in images x and y is respectively. Where represents the covariance of brightness between images x and y . C_1 , C_2 are constants utilized for reliable computation.

Three models are tested: WGAN-GP, WGAN-GP with the addition of the squeeze and excitation block (SE-WGAN-GP), and the proposed model RS-WGAN-GP. Table 4 presents a comparison between the proposed model and the other models using samples from the dataset. Three models are compared with each other to decide which has good performance. For WGAN-GP, the PSNR is 23.12; SE-WGAN-GP achieves 25.75, and for the proposed model, RS-WGAN-GP achieves a high PSNR value of 26.33 for using the same image. According to the value shown in Table 3 and the quality of images in Fig. 11, the proposed model RS-WGAN-GP achieves the best performance compared to the other models.

After applying the augmentation technique by using the proposed model RS-WGAN-GP to balance between classes. The dataset consists of 128 and 119 images for Mpox and others, respectively. Then the number of features extracted by using Vision Transformer, handcrafted, VGG16, VGG19, and ResNet 50 is 768, 210, 512, 512, and 2048, respectively. To enhance the proposed model, each extracted feature is evaluated individually and then combined one by one to improve classification performance. Table 5 presents samples of the MSLD dataset within the proposed model. The table contains the row image, the generated image using GAN, the number of extracted features and their types, and the final selected features. As shown in Table 6, the performance of classification is increasing after combining more features, and the best classifiers used are SVM and random forest, achieving an accuracy of 87.84% and 85.13%, respectively. Then all combined features have a size of (245, 4050), while 245 represents the number of images and 4050 is the number of extracted features.

For reduction, the proposed MB-GWO model is used for reducing the feature. The number of epochs is set to 30; the number of wolves is 10. SVM is used as the classifier in the objective function. The dataset is split into 30% testing and 70% training. The reduced features are evaluated using various classifiers, and the performance of the model is measured using confusion metrics, including accuracy, precision, recall, F1 score, and specificity. As shown in Table 7, after multiple trials, the highest accuracy is achieved with the SVM classifier at 90.54% with a 0.0949 fitness function. Across more trials, the SVM demonstrates higher classification accuracy.

To assess the statistical effectiveness of the proposed MB-GWO feature selection method, a Wilcoxon signed-rank test was conducted to compare the classifier's performance using the full feature set versus the selected subset. The test yielded a Wilcoxon statistic of $W = 45.5000$ with a corresponding p -value of 0.0047. Since the p -value falls below the conventional significance threshold of 0.05, the results indicate a statistically significant difference in performance, thereby validating the contribution of the proposed MB-GWO feature selection, which contributes meaningfully to improving classification performance.

As illustrated in Fig. 12, confusion matrices are presented for the multi-classifier that achieved a lower fitness function value of 0.0988. Table 8 introduces the performance metric for five classifiers, highlighting that the SVM achieves high performance compared to others. The accuracy, precision, F1-score, and specificity achieved by SVM are 90.54%, 91.43%, 90.14%, and 90.50%, respectively.

As shown in Fig. 13, it presents the LIME chart, which provides the score of the 50 most important features that affect the classification process in the models. The figure provides various features such as GLCM, which is named for

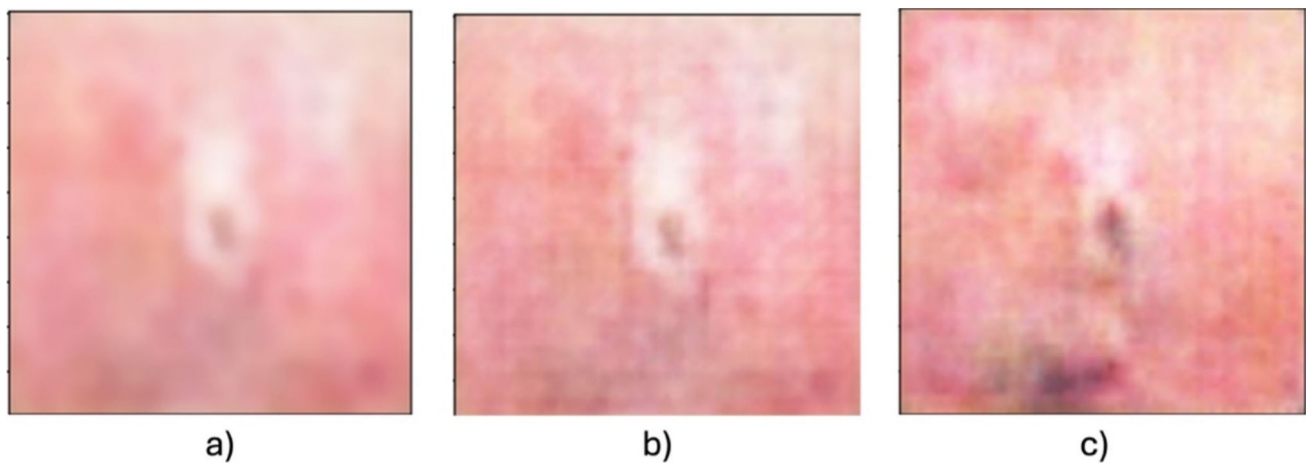


Fig. 11 Samples from three model where **a** is the proposed model, **b** refer to the SE-WGAN, and **C**) is the standard WGAN-GP

Table 5 Samples of the MSLD dataset within the proposed model

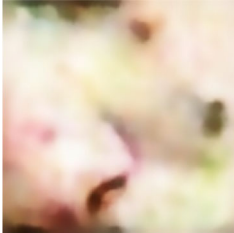
	Sample 1	Sample 2	Sample 3	Sample 4
Original images				
RS-WGAN-GP				
Extracted Features	Handcrafted = 210, VGG16 = 512, VGG19 = 512, ViT = 768, ResNet50 = 2048			
Combined Features	Number of features 4050			
MB-GWO	2101			

Table 6 Comparison on different combination of feature extraction

Model	RF	SVM	NB	KNN	LR
Hand feature	74.32%	56.76%	71.62%	54.05%	63.51%
Hand feature + Vision T	77.03%	58.11%	74.32%	67.57%	85.13%
Hand feature + Vision T. + VGG16	78.38%	83.78%	71.62%	75.67%	72.97%
Hand feature + Vision T. + VGG16 + VGG19	78.38%	85.14%	75.68%	74.32%	74.32%
Handcrafted feature + Vision T. + VGG16 + VGG19 + ResNet 50 (The proposed)	85.13%	87.84%	78.38%	77.03%	75.68%

Bold indicates best obtained results for the classification process

Table 7 Accuracy results of the best fitness value of different classifiers

Fitness value	RF	SVM	NB	KNN	LR	Selected features
0.0988 (The proposed)	81.081%	90.54%	78.38%	79.73%	82.43%	2101
0.1124	81.08%	89.18%	77.03%	82.43%	79.73%	2288

Bold indicates best obtained results for the classification process

hand features, VGG16, VGG19, ViT, and ResNet features. The horizontal bar represents the importance of the feature inflection, while the vertical bar shows the feature name. For the horizontal bar, the positive value indicates an increase in the prediction score, while the negative value indicates a decrease. According to this, the graph shows that VGG19 and ResNet are the most influential in the prediction process.

The classification of MSID dataset

The proposed model applies to the MSID, which has four classes: chickenpox, measles, Mpox, and normal. RS-WGAN-GP is used for generating images. The model is trained for 1800 epochs, and after the training process, 64 images are generated. The generated images contain some noise and blurriness, so a bilateral filter is applied to enhance the synthetic data. The quality of the generated data is evaluated using PSNR and SSIM. Figure 14 represents samples of the generated images with PSNR and SSIM values.

The number of features extracted using Vision Transformer, handcrafted, VGG16, VGG19, and ResNet 50 is 768, 210, 512, 512, and 2048, respectively. To improve the proposed model, each extracted feature is tested separately and combined one by one to improve classification. Table 9 presents samples of the MSID dataset within the proposed model. The table contains the row image, the generated image using GAN, the number of extracted features and their types, and the final selected features. As shown in Table 10, the classification performance has been increased after combining more features, and the best classifiers used are random forest and SVM, achieving an accuracy of 89.74% and 86.75%, respectively. Then all combined features have a size of (779, 4050), while 779 represents the number of images and 4050 is the number of extracted features.

For reduction, the proposed MB-GWO model is used for reducing the feature. The number of epochs used is 30; the number of wolves is 10. The classifier used in the objective function is SVM. The data is split into 30% testing and 70% training. The reduced features are tested with different classifiers, and the performance of the model is measured by using confusion metrics and by calculating the accuracy, precision, recall, F1 score, and specificity. As shown in Table 11, after more trials, the best accuracy has been achieved with the RF

classifier at 91.88% with a 0.1239 fitness function. For more trials, the Random Forest has higher classification accuracy.

As illustrated in Fig. 15, confusion matrices are shown for the multi-classifier, which has a lower fitness function at 0.1239. Table 12 introduces the performance metric for five classifiers that show the Random Forest achieves high performance compared to other classifiers, where accuracy, precision, recall, F1-score, and specificity have 91.88%, 92.21%, 91.88%, 91.62%, and 96.93%, respectively.

As shown in Fig. 16, it presents the LIME chart, which provides the score of the 50 most important features that affect the classification process in the models. The figure provides the various features as GLCM, which is named for hand features, VGG16, VGG19, ViT, and ResNet features. The horizontal bar represents the importance of the feature inflection, while the vertical bar shows the feature name. For the horizontal bar, the positive value indicates an increase in the prediction score, while the negative value indicates a decrease. According to this, the graph shows that VGG16 and ViT are the most influential in the prediction process of the MSID dataset.

In order to ensure the effectiveness of the proposed methodology, Table 13 presents a comparison of the proposed model with recent related research. All mentioned studies utilized the same datasets, but they used different class types: binary and multiple classifications. The proposed model in study two, which used four classes, achieved the highest accuracy compared to other studies. In the binary classification study, the proposed model attained an accuracy of 90.54%, whereas the model by Sahin et al. achieved a slightly higher accuracy of 91.11%. However, despite its higher accuracy, Sahin et al.'s model demonstrated lower precision (90%), specificity (90%), and F1-score (90%), indicating certain limitations. In contrast, the proposed model in the first study, designed for binary classification, exhibited slightly lower accuracy but outperformed in precision (91.43%), specificity (90.50%), and F1-score (90.14%), making it a more balanced and reliable classification model.

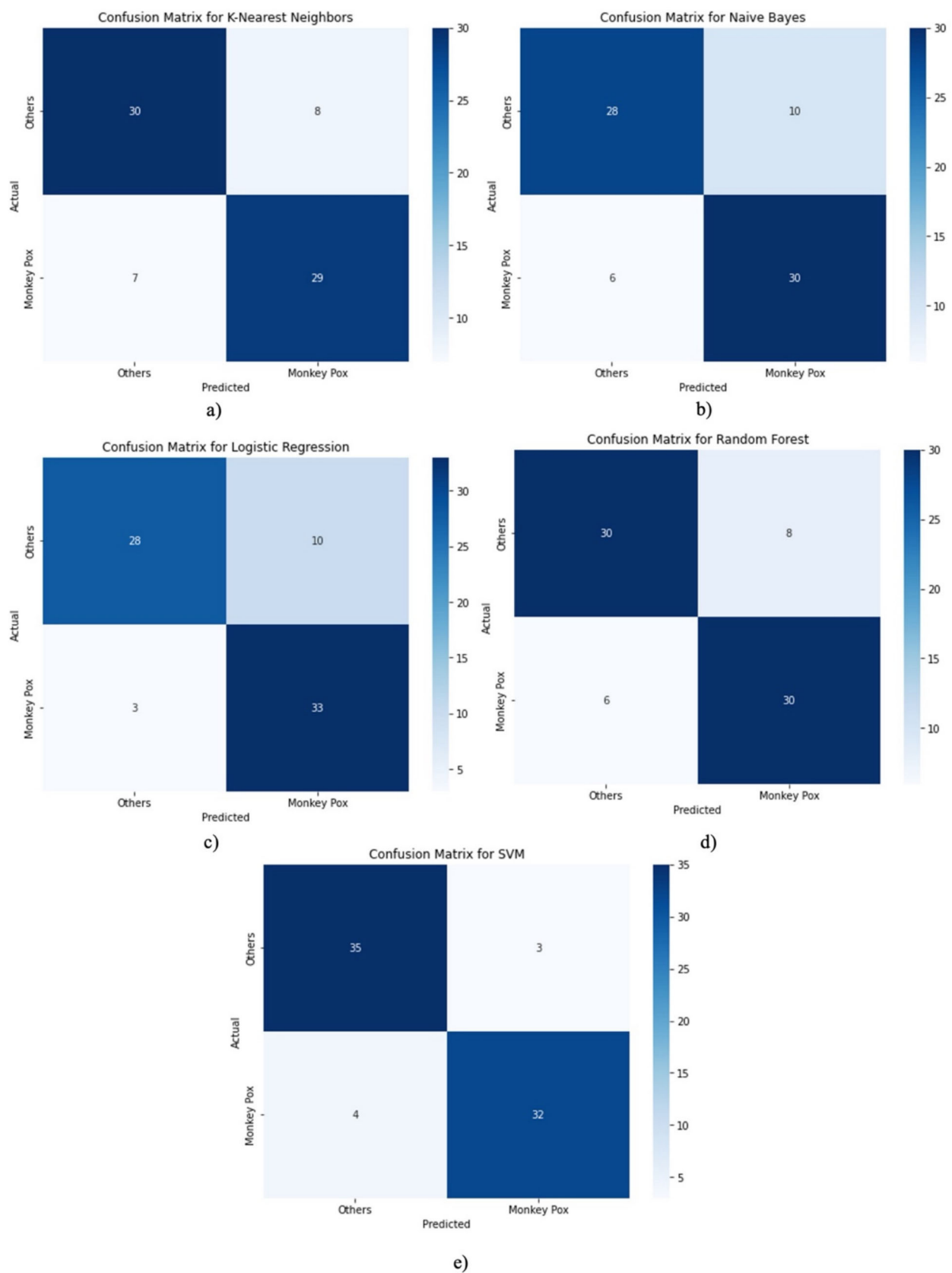


Fig. 12 Confusion matrices for multi-class classifiers at a fitness value of 0.09, where: **a** represents the KNN classifier, **b** the Naïve Bayes classifier, **c** logistic regression, **d** random forest, and **e** the SVM classifier, which achieves the highest performance

Table 8 Performance metrics for the proposed model, where the fitness value is 0.0988

Metrics	RF	SVM	NB	KNN	LR
Accuracy	81.081%	90.54%	78.38%	79.73%	82.43%
Precision	78.95%	91.43%	75%	78.38%	76.74%
Recall	83.33%	88.89%	83.33%	80.56%	91.67%
F1-Score	81.08%	90.14%	78.95%	79.45%	83.54%
Specificity	81.14%	90.50%	78.51%	79.75%	82.68%

Bold indicates best obtained results for the classification process

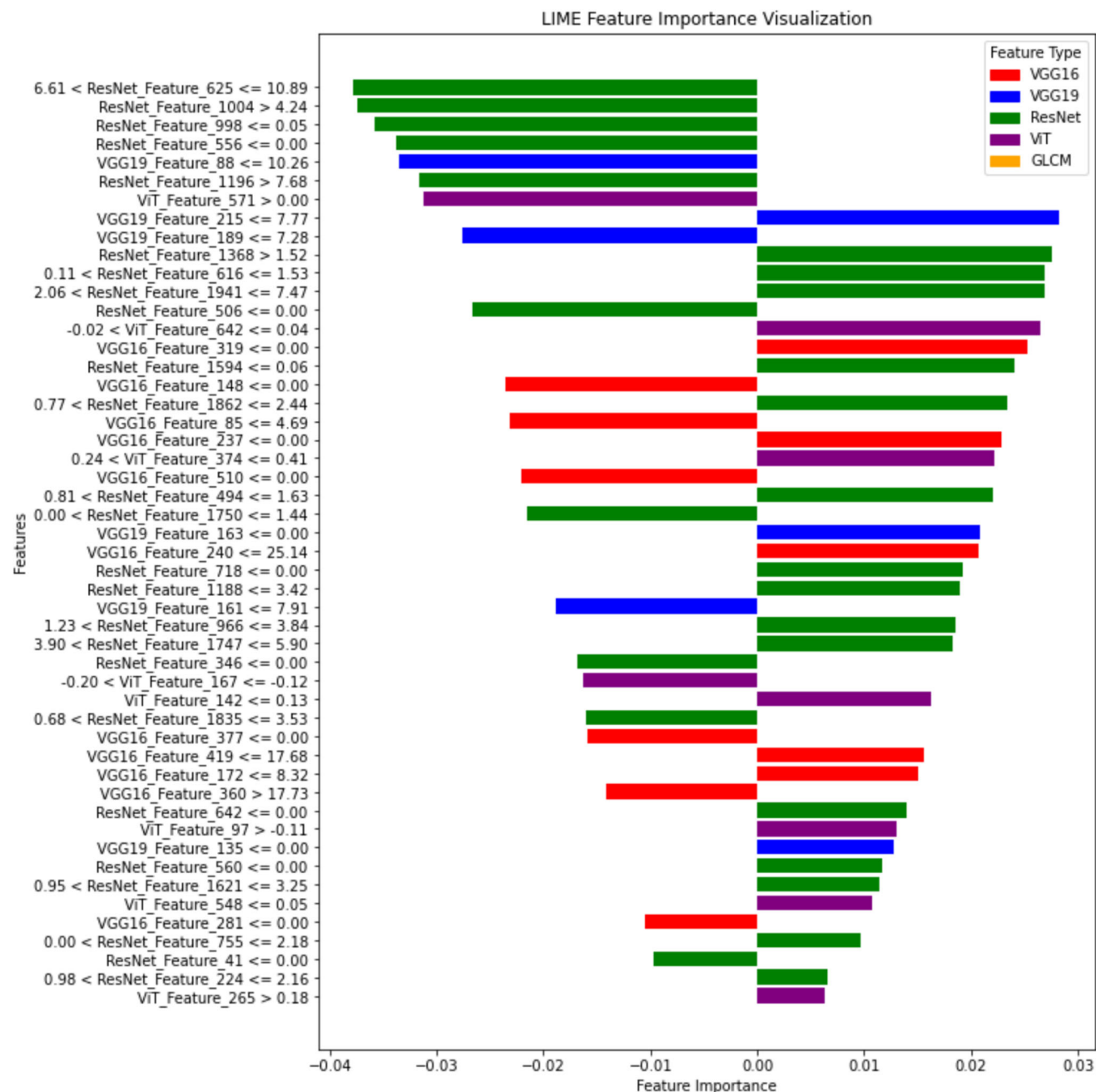
**Fig. 13** The LIME explanation of the selected features of classification MSLD



Fig. 14 Samples from generated images of MSID

Table 9 Samples the MSID dataset alongside the proposed model

	Sample 1	Sample 2	Sample 3	Sample 4
Original images				
RS-WGAN-GP				
Extracted Features	Handcrafted = 210, VGG16 = 512, VGG19 = 512, ViT = 768, ResNet50 = 2048			
Combined Features	Number of features 4050			
MB-GWO	2210			

Table 10 Accuracy for multi-classifier for testing features combination

For adding 779	RF	SVM	NB	KNN	LR
Hand feature	66.24%	48.29%	51.28%	39.3%	58.55%
Hand feature + Vision T	89.74%	48.72%	83.76%	62.82%	73.93%
Hand feature + Vision T. + VGG16	89.32%	83.76%	85.04%	79.49%	84.19%
Hand feature + Vision T. + VGG16 + VGG19	88.03%	84.62%	84.62%	79.91%	86.32%
Hand feature + Vision T. + VGG16 + VGG19 + ResNet 50 (The proposed)	89.74%	86.75%	84.62%	82.05%	86.32%

Bold indicates best obtained results for the classification process

Table 11 Accuracy results of the best fitness value of different classifiers

Fitness value	RF	SVM	NB	KNN	LR	Selected features
0.1239 (The proposed)	91.88%	88.03%	82.91%	81.62%	85.04%	2210
0.1238	90.60%	88.03%	84.19%	79.91%	85.47%	2199
0.1282	89.74%	87.61%	83.76%	79.49%	84.19%	2271

Bold indicates best obtained results for the classification process

Conclusion

Monkeypox is a rare viral disease caused by the monkeypox virus. Rapid or early detection of monkeypox is one of the most important methods used for prevention and rapid recovery from it. This study proposed a novel multi-stage classification framework, MSCADMpox, that effectively integrates deep learning features and handcrafted features for accurate monkeypox identification. The proposed model was evaluated on two datasets: MSLD and MSID. The proposed framework introduces several key innovations. First, using the RSWGAN model for image enhancement created high-quality synthetic images, shown by better PSNR (26.33) and SSIM (0.61) scores than other image enhancement methods. Second, the combination of pre-trained deep learning models (such as ViT, VGG16, VGG19, and ResNet50) with handcrafted features (such as GLCM, LBP, and HOG) managed to extract both global and local patterns from skin lesion images using the hybrid method for feature extraction. The progressive feature integration strategy ensured that only informative features contributed to the final model. Further-

more, the use of the Modified Binary Grey Wolf Optimizer (MBGWO) for feature selection significantly reduced dimensionality while preserving critical information. A Wilcoxon signed-rank test showed that this proposed model was statistically effective, with a p-value of 0.0047 ($W = 45.5$), meaning it performed significantly better than using all the features. Finally, the MSCADMpox model achieved 90.54% accuracy on the MSLD dataset and 91.88% on the MSID dataset. These results demonstrate the effectiveness of combining synthetic data generation, diverse feature extraction (deep and handcrafted), and metaheuristic feature selection. Unlike prior models that rely on a single feature type, MSCADMpox offers a more comprehensive and adaptable approach. As part of future work, we intend to combine multiple datasets to increase the volume and diversity of training images, explore real-time deployment of the model in practical applications such as mobile applications or clinical settings, and apply deep learning models to further enhance classification performance.

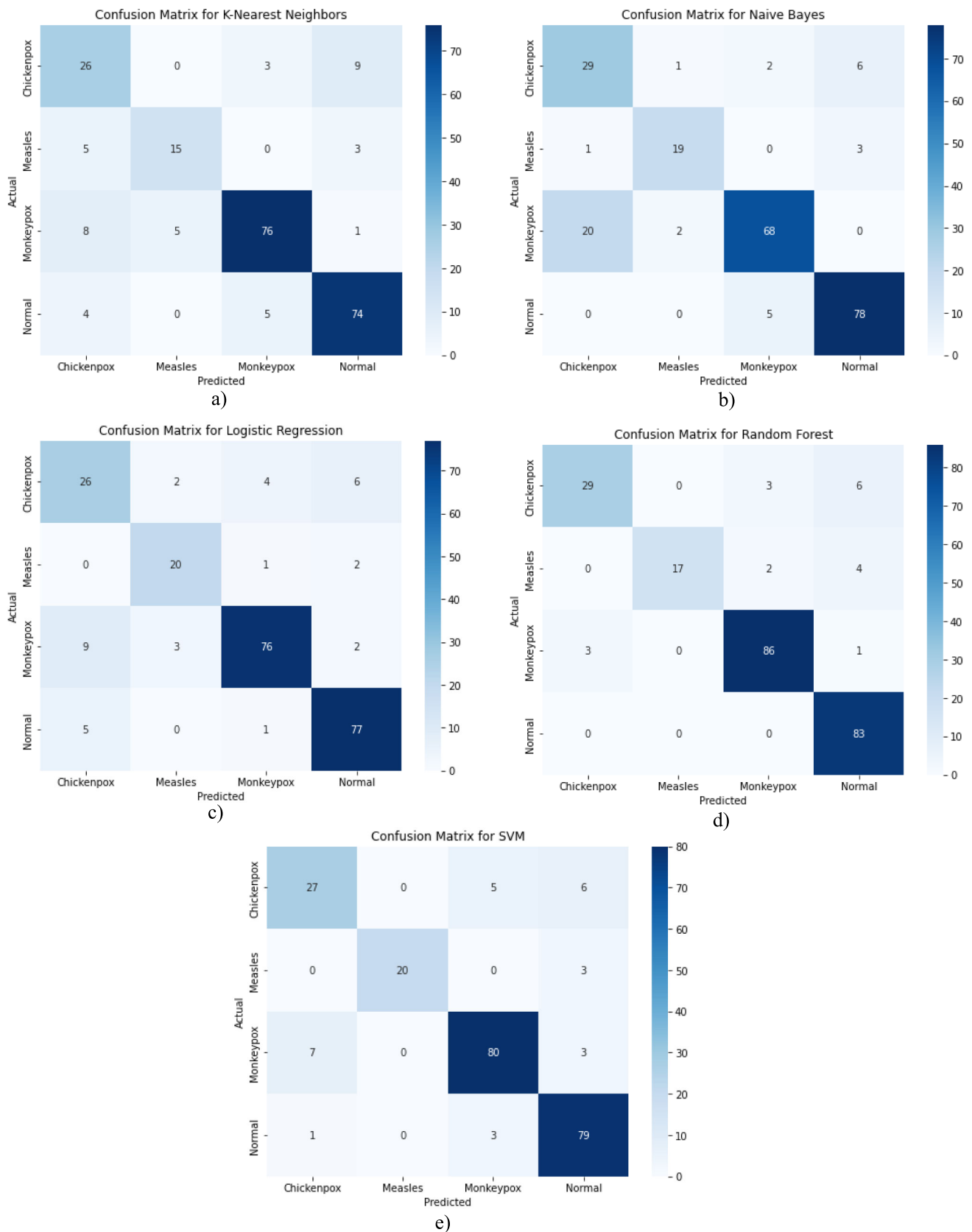


Fig. 15 Confusion matrices for multi-classifiers at a fitness value is 0.1239, where: **a** represents the KNN classifier, **b** the Naïve Bayes classifier, **c** logistic regression, **d** random forest, and **e** the SVM classifier, which achieves the highest performance

Table 12 Performance metrics for the proposed model, where the fitness value is 0.1239

Metrics	RF	SVM	NB	KNN	LR
Accuracy	91.88%	88.03%	82.91%	81.62%	85.04%
Precision	92.21%	88.11%	84.58%	82.16%	85.46%
Recall	91.88%	88.03%	82.91%	81.62%	85.04%
F1-Score	91.62%	87.94%	83.25%	81.76%	85.14%
Specificity	96.93%	95.60%	94.26%	93.69%	94.92%

Bold indicates best obtained results for the classification process

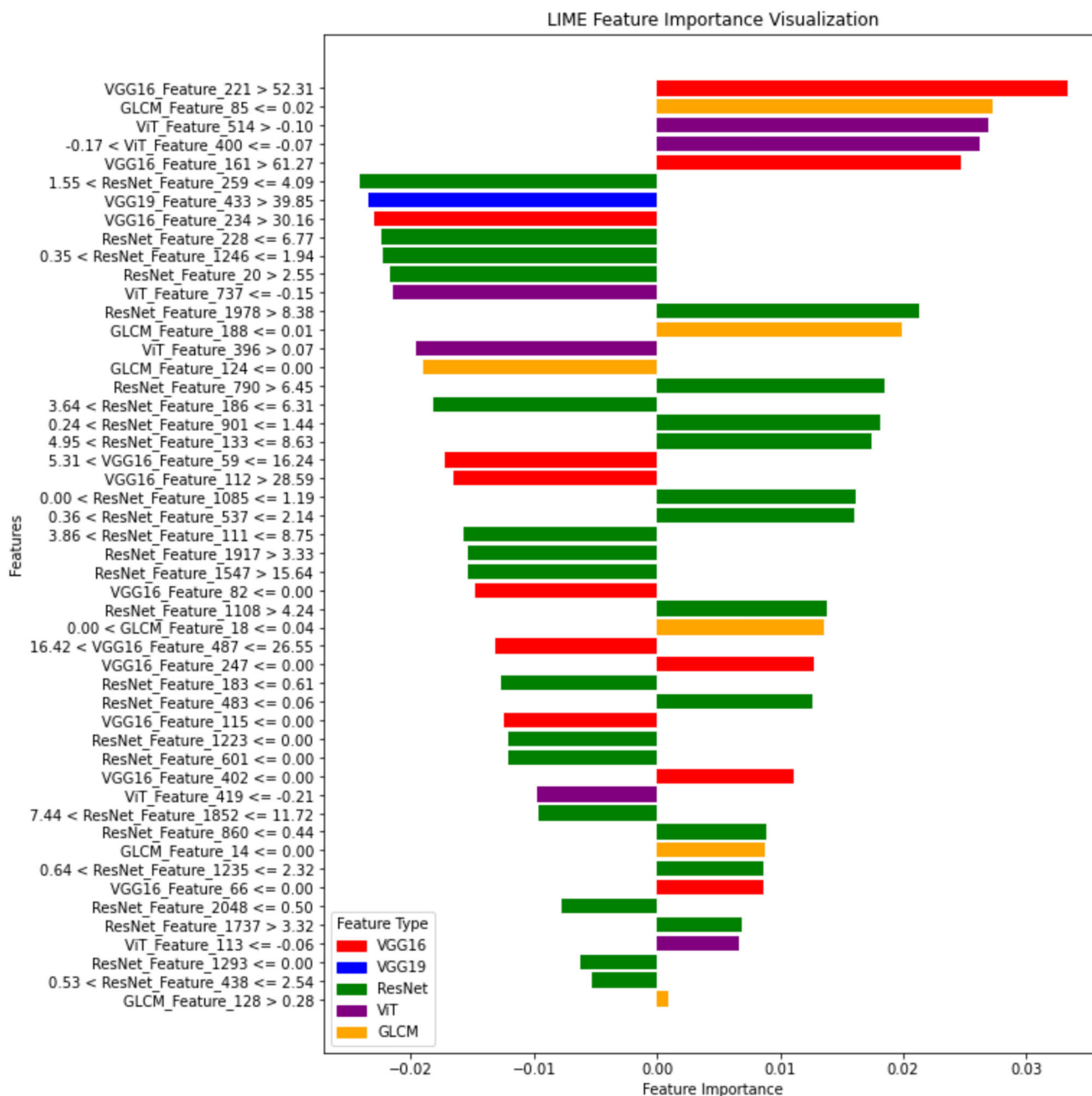
**Fig. 16** The LIME explanation of the selected features of classification MSID

Table 13 Comparative results with other studies

Ref	Dataset	classes	Model	Accuracy	Specificity	Precision	Recall	F1-score
Uysal et al. [11]	MSID	4	CNN + LSTM	87%	–	93%	87%	90%
Irmak et al. [12]	MSID	4	MobileNetV2	91.38%	–	90.5%	86.75%	88.25%
Sahin et al. [15]	MSLD	2	MobileNetv2	91.11%	90%	90%	–	90%
Ali et al. [39]	MSLD	2	ResNet50	82.96%	–	87%	83%	84%
The Proposed	MSLD	2	Proposed study one	90.54%	90.5%	91.43%	88.89%	90.14%
The Proposed	MSID	4	Proposed study two	91.88%	96.93%	92.21%	91.88%	91.62%

Bold indicates best obtained results for the classification process

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Data availability The used datasets: the Monkeypox Skin Lesion Dataset (MSLD) is available in [18] and the Monkeypox Skin Image Dataset (MSID) is available in Ref [19].

Declarations

Conflict of interest The authors declare that they have no competing interests nor conflict of interests for the current study.

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